

Some review questions - 9th week

- Q1** Given a matrix $A \in \mathbb{R}^{m \times n}$, prove that if A has linearly independent columns, then $A^\top A$ is invertible. (Hint: You can use the result " $N(A^\top A) = N(A)$ " and "if $\forall u \in \mathbb{R}^n, u^\top x = 0$, then $x = 0$ ".
- Q2** Given a vector $b \in \mathbb{R}^m$ and a matrix $A \in \mathbb{R}^{m \times n}$ with linearly independent columns, write the solution of the least squares problem $\min_{x \in \mathbb{R}^n} \|Ax - b\|^2$. You can directly use the result that $A^\top A$ is invertible when A has independent columns.
- Q3** Given a vector $b \in \mathbb{R}^m$ and a matrix $A \in \mathbb{R}^{m \times n}$, show that the equation $A^\top Ax = A^\top b$ admits at least one solution, using the fact that $N(A^\top A) = N(A)$.
- Q4** Give the definitions of orthogonal sets of vectors and orthonormal sets of vectors.

Q5 Prove that any orthogonal set of two nonzero vectors is linearly independent. [Remark: This result holds for any orthogonal set, but we only ask you to prove the case where the set has two vectors.]

Q6 Give the definition of an orthonormal basis.

Q7 State a necessary and sufficient condition, involving a matrix product, for a matrix to be orthogonal.

Q8 Given an orthogonal matrix $Q \in \mathbb{R}^{n \times n}$, express Q^{-1} in a simple form.

Q9 Let $Q \in \mathbb{R}^{n \times n}$ be an orthogonal matrix, and let $x, y \in \mathbb{R}^n$. What can be said about $\langle Qx, Qy \rangle$ and $\|Qx\|$? Prove one of these two results using your knowledge of orthogonal matrices.

Q10 State the result on the representation of elements of a linear space with respect to a basis.

Q11 Given $U = \{u_1, \dots, u_n\}$, an orthonormal basis of $\mathbb{R}^n$, express a vector $v \in \mathbb{R}^n$ as a linear combination of $u_1, \dots, u_n$ (give an explicit expression for the coefficients). Prove the result.

Q12 Given a subspace $S \subset \mathbb{R}^n$ and a vector $b \in \mathbb{R}^n$, give the definition of the projection p of b onto S .

Q13 Given a subspace $W \subset \mathbb{R}^n$, an orthonormal basis $\mathcal{B} = \{w_1, \dots, w_k\}$ of W , and a vector $v \in \mathbb{R}^n$, express the projection p of v onto W in terms of $w_1, \dots, w_k$.

Q14 Describe the Gram-Schmidt procedure to construct an orthonormal basis $\mathcal{V} = \{v_1, \dots, v_k\}$ from a basis $\mathcal{U} = \{u_1, \dots, u_k\}$ such that for all $i \in \{1, \dots, k\}$, $\text{span}\{v_1, \dots, v_i\} = \text{span}\{u_1, \dots, u_i\}$.

Q15 Use Gram-Schmidt process to compute an orthonormal basis of $\text{span}\{u_1, u_2, u_3\}$, where $u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $u_2 =$

$$\begin{bmatrix} -1 \\ 4 \\ 4 \\ -1 \end{bmatrix}, \text{ and } u_3 = \begin{bmatrix} 4 \\ -2 \\ 2 \\ 0 \end{bmatrix}.$$

Q16 Give the formula for the determinant of a matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, for $a, b, c, d \in \mathbb{R}$. Give a geometric interpretation of this determinant in terms of area.

Q17 Let a_1, a_2 , and a_3 be three vectors in $\mathbb{R}^3$, and consider $A = (a_1, a_2, a_3) \in \mathbb{R}^{3 \times 3}$. Give a geometric interpretation of $\det(A)$. State the condition on $\det(A)$ for A to be invertible.